

Biological Classification

SHORT NOTES

Table 1: Characteristics of Five Kingdoms proposed by Whittaker (1969)

Characters	Five Kingdoms				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present with chitin	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular/ loose tissue	Tissue/organ	Tissue/organ/ organ system
Mode of nutrition	Autotrophic (Chemosynthetic and photosynthetic) and Heterotrophic (sapro-phytic/ parasitic)	Autotrophic (Photosynthetic) and Hetero-trophic	Heterotrophic (Saprophytic/ Parasitic)	Autotrophic (Photosyn-thetic)	Heterotrophic (Holozoic/ Saprophytic etc.)

Kingdom Monera

- ❖ Has bacteria a sole member.
- ❖ Bacteria can have shapes like: Coccus (sphere cal). Bacillus (rod-shaped). Vibrio (comma shaped) and sprillum (spiral shaped)
- ❖ Bacteria found almost everywhere and can be Photosynthetic autotrophs, Chemosynthetic autotrophs or Heterotrophs.

Archaeobacteria

- ❖ Halophiles (salt-loving)
- ❖ Thermoacidophiles (in hot springs)
- ❖ Methanogens (in marsh and in gut of ruminant animals produce methane gas.)

Eubacteria

- ❖ Photosynthetic autotrophs like Cyanobacteria. some like *Anabaena* have specialised cells called heterocysts for nitrogen fixation.
- ❖ **Chemosynthetic autotrophs:** Oxidise various inorganic substances like nitrates/nitrites, ammonia and use released energy for their ATP production.
- ❖ **Heterotrophic bacteria:** decomposes, help in making curd, production of antibiotics, N_2 fixation, cause diseases like cholera, typhoid.

Mycoplasma: Completely lack cell wall. smallest living cells. Can survive without oxygen. Pathogenic in animals and plants.

Kingdom Protista: (All single celled eukaryotes)

Forms a link between plants, animals and fungi.

(i) **Chrysophytes:** (Has diatoms and golden algae)

- ❖ Cell walls have silica and cell walls overlap to fit together like a soap box.
- ❖ Their accumulation forms 'Diatomaceous Earth'.
- ❖ Used in polishing, filtration of oils and syrups.

(ii) **Dinoflagellates:** Marine, photosynthetic, cell wall has cellulose. Two flagella "one longitudinal and other transversely in a furrow between wall plates.

(iii) **Euglenoids:** Have protein rich layer 'pellicle' which makes body flexible. Photosynthetic in presence of sunlight but become heterotrophs if they do not get sunlight.

(iv) **Slime Moulds:** Saprophytic protists form aggregates to form plasmodium grows on decaying twigs and leaves. Spores have true walls which are extremely resistant and survive for many years.

(v) **Protozoans:**

- ❖ **Amoeboid:** Catch prey using pseudopodia, e.g., Amoeba.
- ❖ **Flagellated:** More flagella. Cause disease like sleeping sickness e.g., *Trypanosoma*.
- ❖ **Ciliated:** Have cilia to move food into gullet and help in locomotion. e.g., *Paramecium*.
- ❖ **Sporozoonans:** Have infective spore like stage in life cycle, e.g., Plasmodium which causes

Kingdom Fungi

- ❖ Non chlorophyllous hyphae
- ❖ Network of hyphae called mycelium
- ❖ Cell wall of chitin and polysaccharides
- ❖ Grow in warm and humid places
- ❖ Saprophytic, parasitic, symbiotic (Lichen) e.g., *Puccinia* (rust causing), *Penicillium*.

Table 2: Characteristics of Four Classes of fungi

Classes	Typical Examples	Sexual Reproduction	Asexual Reproduction	Hyphae
Phycomycetes	Rhizopus, Mucor, Albugo	Zygospores	Zoospores (motile) Aplanospores (non-motile)	Aseptate and Coenocytic
Ascomycetes (Sac fungi)	Penicillium (multicellular), yeast (unicellular), Aspergillus, Claviceps, Neurospora, morels and truffles (truffles)	Ascospores produced endogenously in sac like asci	Conidia produced on conidiophore	Branched and septate
Basidiomycetes (Club fungi)	Mushrooms (Agaricus) smut (Ustilago), Rust (Puccinia) Bracket fungi or Puff balls.	Basidiospores produce exogenously on the basidium	Generally asexual spores are not found but reproduce vegetatively by fragmentation	Branched and septate
Deuteromycetes (Imperfect fungi)	Alternaria, Colletotrichum. Trichoderma	Sexual phase has not been observed	Conidia	Branched and Septate

Classes of Fungi

(i) **Phycomycetes:**

- ❖ Grow on decaying wood
- ❖ Mycelium Aseptate
- ❖ spores produced endogenously
- ❖ Asexual reproduction by Zoospores or Aplanospores e.g., *Rhizopus*, *Albugo*.

(ii) **Ascomycetes:**

- ❖ Also known as 'sac fungi'
- ❖ Mycelium branched and septate
- ❖ Spores: Asexual spores are called conidia produced exogenously on the conidiophores.

- ❖ Sexual spores are called ascospores produced endogenously in ascus produced inside fruiting body called ascocarp. e.g., *Aspergillus*, *Neurospora*.

(iii) **Basidiomycetes:**

- ❖ Mycelium septate.
- ❖ The asexual spores are generally not found but vegetative reproduction by fragmentation is common.
- ❖ The basidiospores are exogenously produced on the basidium (pl. basidial).
- ❖ The basidia are arranged in fruiting bodies called basidiocarps.
- ❖ Some common members are *Agaricus* (mushroom), *Ustilago* (smut) and *Puccinia* (rust fungus).

(iv) Deuteromycetes:

- ❖ Called as 'Fungi imperfecti' as sexual form(perfect stage) is not known for them.
- ❖ Once perfect (sexual) stages of members of dueteromycetes were discovered they were often moved to ascomycetes and basidiomycetes.
- ❖ The deuteromycetes reproduce only by asexual spores known as conidia.
- ❖ The mycelium is septate and branched.
- ❖ Some members are saprophytes or parasites while a large number of them are decomposers of litter and help in mineral cycling.
- ❖ Some examples are *Alternaria*, *Colletotrichum* and *Trichoderma*.

Viruses

- ❖ They did not find a place i classification. Take over the machinery of host cell on entering it but as such they have inert crystalline structure. So, difficult it call them living or **non-living**.
- ❖ Pasteur gave the term 'Virus' i.e., **poisonous fluid**.
- ❖ **D.J. Ivanowsky** found out that certain microbes cause
- ❖ **M.W. Beijerinck** called fluid as '**Contagium vivum fluidum**' as extracts of infected plants of tobacco could cause infection in healthy plants.
- ❖ **W.M. Stanely** showed viruses could be arystalli sed to form cry stals of protein which are inert outside their specific host.

Structure of Virus

- ❖ Its a nucleoprotein Covered by protein called **Capsid**. Capsid is made up of capsomeres arranged in helical or polygeometric forms.
- ❖ Have either DNA or RNA as genetic material which may be single or double stranded.
- ❖ Usually plant viruses have single stranded RNA
- ❖ Bacteriophages have double stranded DNA and animal virus have single or double stranded RNA or double stranded DNA.

Diseases caused: Mumps, Small pox, AIDA etc.

Viroids

- ❖ Infectious agent, free RNA (lack protein coat) of low molecular weight
- ❖ Causes **potato spindle tuber disease**.
- ❖ Discovered by **T.O. Diener**.

Prions

- ❖ They consist of **abnormally folded protein** without genetic material.
- ❖ They cause mad cow disease in cattle and **Cr-Jacob disease (CJD)** in humans.

Lichens

- ❖ Symbiotic association between algal component (**Phycobiont**) and fungal component (**mycobiont**). Algae provide food. Fungi provide shelter and absorb nutrients for alga.
- ❖ Good pollution indicators as they do not grow in polluted area.
- ❖ Mycorrhiza: Symbiotic association between fungi and roots of higher plants.